

Alicia Grechu

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EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA, Los Angeles, CA
Bachelor of Science, Computer Engineering and Computer Science

Expected Grad: May 2028
Video Game Programming Minor

SKILLS

C++, C#, Python, Java, Verilog | Unreal Engine (C++ & Blueprints), Unity | Perforce, Git/GitHub | Agile/Scrum

WORK EXPERIENCE

Fear Elements Studio, Remote Internship

Jun. 2026-Present

Gameplay Programmer & Programming Lead

- **Own** the player movement, health, and HUD systems in **Unreal Engine (C++ and Blueprints)**, including 3 penguin locomotion modes (waddling, swimming, tobogganing) with distinct physics and state transitions
- Lead technical direction for a 4-6 person **gameplay programming team**, setting coding standards, reviewing pull requests, and breaking down feature work into manageable tasks and sprints
- Balanced hands-on implementation of gameplay features with mentoring fellow programmers and **coordinating across design and production** to keep development on schedule

RELEVANT PROJECTS

Crease (Gameplay Engineer - Team Project) | Unity/C#, Perforce

Jun. 2026-Present

- **Own** the WindSplineTube system in **Unity/C#**, which guides the player along 3D splines using tangent-based flow and a radial centering force, with reversible direction and particle VFX
- Iterated on the centering force through 3 rounds of playtesting to balance guidance with player control, so players are guided by the wind but can always escape the current if desired

Frogger (Gameplay Programmer - Solo Project) | C++, GitHub

Sep. 2026

- Built a Frogger clone from scratch in **C++ with no engine**, on a custom framework developed across multiple games: a hybrid actor/component model (inheritance plus component composition), actor lifecycle management with destruction deferred until after the update loop, and a draw-order sprite renderer with a texture cache
- Implemented AABB collision with minimum-overlap resolution, verified with unit tests on GitHub Actions, plus a text-file level loader that spawns actors from a tile grid

Large Language Mimic (Gameplay Programmer - Team Project) | Unity/C#, GitHub

Jan. 2026-May 2026

- Programmed core gameplay mechanics, NPC behavior, and UI feedback in **Unity/C#** for a satirical 2D game where players impersonate an AI chatbot, with branching dialogue and multiple endings
- Implemented character animations and scene sequencing that hands off cleanly between minigames (e.g. file redaction, photo cleanup), integrating gameplay systems across a 25-person programming team

ACTIVITIES & INVOLVEMENT

USC Open Alpha - Game Programmer & Vice President of Tech

Sep. 2025-Present

- Lead technical direction for the club's semesterly **Unity/C#** game as VP of Tech: set up and maintain the GitHub repository, project configs, and builds, and manage releases on itch.io and Steam
- Developed and teach a **Unity/C#** curriculum to onboard new programmers to the team
- Previously shipped 2 games as a gameplay programmer; work with interdisciplinary teams (design, programming, art, sound) to design mechanics, implement features, conduct playtests, and iterate fast based on feedback

USC Makers - Software Engineer

Sep. 2025-May 2026

- Built FIFO, a smart fridge attachment that tracks inventory and expiration dates using sensors, barcode scanning, and **Python computer vision**, paired with a **React Native app database** and **Unity/C#** companion app
- Developed the **Unity/C#** companion app's calendar tracking and avatar customization systems, and integrated them with the React Native app's database through a **C#** sync layer so changes made in-game update the shared database